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Clinical study

Thromboembolic complications following intravenous immunoglobulin therapy in immune-mediated neurological disorders

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Background: Minor adverse events of intravenous immunoglobulin (IVIg) include flu-like symptoms, eczematous skin reaction, electrolyte disturbance, and transient leukopenia. On rare occasions, serious complications such as aseptic meningitis, arrhythmia, decrease in blood pressure, and thromboembolic complications (TEC) have been described. The current study aimed to understand the frequency and clinical features of TEC related to IVIg administration in patients with immune-mediated neurological disorders.
Methods: We conducted a retrospective chart review of hospitalized patients with immune-mediated neuromuscular or neuroimmunological disorders treated with IVIg from January 2018 to March 2020 in a single tertiary hospital.
Results: During the study period, 61 patients were treated with a total of 364 IVIg infusions over 84 treatment courses. Among them, we identified 3 TEC cases that occurred during or after the completion of IVIg therapy: two patients with myasthenia gravis (F/60 and F/80) and one patient with Guillain-Barré syndrome (F/79) had undergone arterial TEC (two for ischemic stroke and one for pulmonary thromboembolism). The rates of TEC per patient, per treatment course, and per infusion were 4.91% (3/61), 3.57% (3/84), and 0.82% (3/364), respectively.
Conclusion: The risk of developing TEC upon receiving IVIg infusions is generally low in patients with immune-mediated neurological disorders; however, IVIg-related TEC should be cautiously monitored for in critically ill elderly patients with vascular risk factors, especially those suffering from myasthenic crisis.

1. Introduction

Intravenous immunoglobulin (IVIg) is a purified plasma fraction of polyspecific immunoglobulin G that is pooled from the human plasma [1]. It has been increasingly used for the treatment of immune-mediated neuromuscular or neuroimmunological disorders, including Guillain-Barré syndrome (GBS), chronic inflammatory demyelinating polyradiculoneuropathy, multifocal motor neuropathy, myasthenia gravis (MG), and autoimmune encephalitis [1,2]. IVIg is generally regarded as a safe medication with few reports on minor adverse reactions that include flu-like symptoms, eczematous skin reaction, electrolyte disturbance, and transient leukopenia. However, serious complications such as aseptic meningitis, arrhythmia, decrease in blood pressure, and thromboembolic complications (TEC) have been rarely described in the context of IVIg infusions [3].

Several previous studies had reported a relatively low incidence of TEC associated with IVIg administration in patients with mainly autoimmune neuromuscular disorders, ranging from 1.2% to 3% [4–6]. More recently, a retrospective study involving 62 patients treated with a total of 616 courses of IVIg indicated that the incidence of early TEC, defined as the occurrence of TEC within 2 weeks from IVIg infusion, was less than 1% per infusion in patients with immune-mediated neuropathy [7]. Those who had cardiovascular diseases or limited mobility were more likely to develop TEC after IVIg therapy [7]. These clinical studies suggested an increased risk of TEC related to IVIg but were limited by a small number of patients. In contrast, a larger study based on the US health insurance records showed that the overall prevalence of stroke in

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patients with primary immunodeficiency who were treated with IVIg was much lower than that in the general population (0.62% vs. 2.6%) [8]. Although there is currently insufficient evidence for the causal relationship between IVIg and TEC, the US Food and Drug Administration issued a warning for the TEC risk of IVIg products in 2013 [9]. Nonetheless, additional clinical and scientific data are required to support the need for preemptive monitoring or primary prevention in patients at high risk of thrombosis who are being prescribed IVIg.

In the present study, we aimed to investigate the frequency and clinical features of TEC associated with IVIg administration in patients with immune-mediated neurological disorders.

2. Methods

We retrospectively reviewed the medical records of hospitalized patients with immune-mediated neuromuscular or neuroimmunological disorders who were treated with IVIg between January 2018 and March 2020 at the Department of Neurology, Inha University Hospital. The patients were treated with either IV-Globulin SN 5%, manufactured by the Green Cross corporation or Liv-Gamma SN 5%, manufactured by SK plasma. The daily dose of IVIg ranged from 400 mg/kg to 1000 mg/kg that was administered for 1–5 days. The initial infusion rate was 0.01 mL/kg/min. If there was no adverse reaction for 30 min, the infusion rate was increased to 0.02 and subsequently to 0.04 mL/kg/min. The maximum infusion rate was set at 0.06 mL/kg/min. For the prevention of allergic reactions, chlorpheniramine (4 mg) was prescribed to the patients, prior to IVIg infusion. The present study was approved by the ethics committee of the Inha University Hospital (IRB No. 2020–08-034). The requirement for informed consent was waived by the IRB because of the retrospective nature of the study.

3. Results

In Table 1, the specific details of IVIg therapy have been outlined. Altogether, 61 patients were treated with a total of 364 IVIg infusions over 84 treatment courses. In Table 2, the clinical information and laboratory findings pertaining to the three patients who manifested IVIg-related TEC, have been depicted.

Table 1
Indications of IVIg therapy.

Final diagnosis	Number of patients	Number of treatment courses	Number of infusions
Guillain-Barré syndrome/ Miller Fisher syndrome	29	30	145
Meningoencephalitis (autoimmune or infectious)	14	17	73
Chronic inflammatory demyelinating polyneuropathy	6	22	71
Myasthenia gravis	6	8	40
Stiff-person syndrome	1	2	10
Acute disseminated encephalomyelitis	1	1	5
Lambert–Eaton myasthenic syndrome	1	1	5
Multifocal motor neuropathy	1	1	5
Inclusion body myositis	1	1	5
ANCA-associated vasculitic neuropathy	1	1	5
Total	61	84	364

3.1. Patient 1

A 60-year-old woman with generalized MG had been experiencing dyspnea and poor oral intake over a few days. She was regularly taking prednisolone 5 mg daily. The subject visited the emergency room in a stuporous state, and an endotracheal intubation was performed. The initial laboratory findings were as follows: creatine kinase (CK) 105 IU/L, CK-MB 16.8 ng/mL, troponin I 3.91 ng/mL, NT-proBNP 25,737 pg/mL, pH 7.247, $PaCO_2$ 68.9 mmHg and, PaO_2 110 mmHg. The electrocardiography revealed an elevation in the ST segment while the transthoracic echocardiogram showed regional wall motion abnormalities in the left anterior descending artery territory and right ventricular apex. A coronary angiography was performed to rule out myocardial infarction wherein minimal coronary artery stenosis was observed. Accordingly, she was diagnosed with stress-induced cardiomyopathy, and supportive management measures were ensured. However, the patient showed no improvement in her mental status as well as the dependence on the mechanical ventilator. A magnetic resonance imaging (MRI) of the brain revealed multiple diffusion-restricted lesions in the left parietal cortex and right posterior inferior cerebellar artery territory. The patient was administered aspirin 100 mg and clopidogrel 75 mg daily, and IVIg 400 mg/kg/day was initiated to rescue the complications associated with MG. Within four days of IVIg infusion, weakness on the right side was noted while dual anti-platelet therapy (DAPT) was maintained. A follow-up brain MRI showed newly developed diffusion-restricted lesions in the left anterior choroidal artery territory and right middle cerebellar peduncle (Fig. 1). The possibility of the development of TEC, associated with IVIg infusion was considered and thus, IVIG therapy was discontinued along with a change from DAPT to warfarin. Further, a daily dosage of prednisolone 60 mg and tacrolimus 3 mg was prescribed to contain the complications associated with MG. Levetiracetam/lacosamide and donepezil were also prescribed for post-stroke epilepsy and post-stroke dementia, respectively. The condition of the patient slowly improved over several weeks and she was finally discharged in a wheelchair-bound state.

3.2. Patient 2

An 80-year-old woman with generalized MG presented with general weakness, binocular diplopia, and dyspnea that developed 10 days earlier. She was not taking any immunosuppressive agents. Upon initial examination, the subject was prescribed prednisolone 20 mg daily and pyridostigmine 60 mg three times a day. Her symptoms gradually improved, but 12 days later, she complained of exacerbated dysphagia and orthopnea. The results from spirometry revealed a forced vital capacity of 29% (of predicted). For the management of the complications associated with MG, the patient was prescribed IVIg at 400 mg/kg/day for five days and the dose of prednisolone was increased to 60 mg daily. Two days after IVIg infusion, the patient suffered a sudden cardiac arrest. Laboratory findings indicated an elevated D-dimer 19.97 ug/ml, while cardiac enzyme was within the normal range. Although chest computed tomography was not performed, hemodynamic instability and D-dimer elevation suggested pulmonary thromboembolism. Subsequently, enoxaparin 40 mg administration was initiated. Electroencephalography showed burst-suppression patterns, and brain MRI findings were compatible with hypoxic-ischemic encephalopathy. Accordingly, phenobarbital and valproic acid were prescribed, but her semi-comatose mentality did not improve. Upon extensive consultation with her family members, a decision was made to suspend the life-saving treatment regimen.

Table 2
Summary of 3 thromboembolic complications.

Sex/Age	Diagnosis	TEC	Onset of TEC	Comorbidities	Laboratory findings
F/60	MG	Ischemic stroke	After IVIg infusion (day 3)	HTN, ischemic stroke	Anti-microsomal antibody (+)
F/80	MG	Pulmonary thromboembolism	2 days after IVIg completion	DM	D-dimer 19.97 ug/mL
F/79	GBS	Ischemic stroke	After IVIg infusion (day 3)	HTN, DM, A-fib	No significant findings

Abbreviations: TEC = thromboembolic complication; MG = myasthenia gravis; GBS = Guillain-Barré syndrome; IVIg = intravenous immunoglobulin; HTN = hypertension; DM = diabetes mellitus; A-fib = atrial fibrillation.

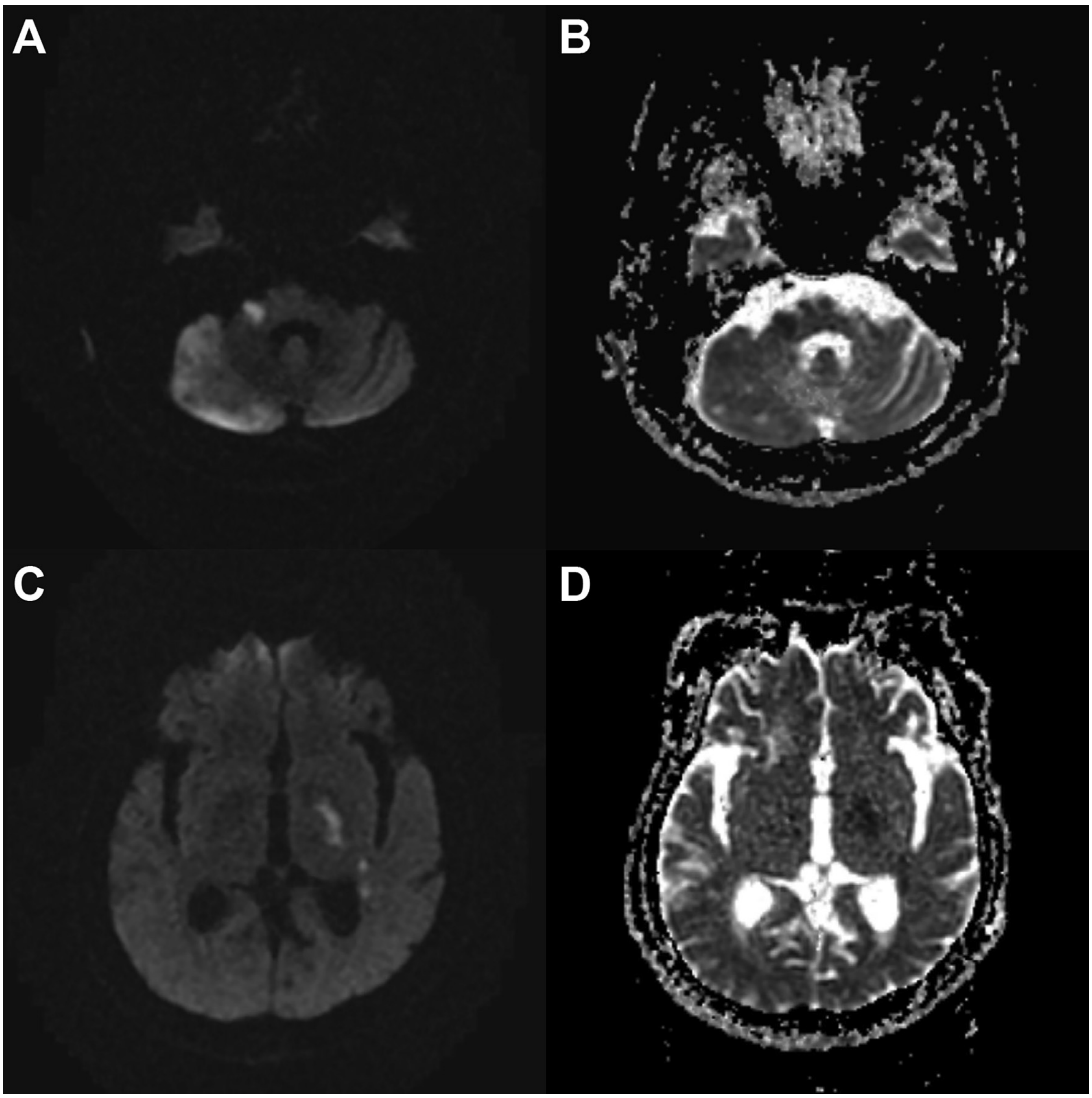

Fig. 1. Brain MRI of patient 1 (F/60, myasthenia gravis). Brain MRI (diffusion weighted image/apparent diffusion coefficient) shows acute infarction in right middle cerebellar peduncle and subacute infarction with normalized apparent diffusion coefficient value in right cerebellum (A and B). Additional axial images show acute infarction in left choroidal artery territory (C and D).

3.3. Patent 3

A 79-year-old woman with a medical history of hypertension, diabetes mellitus, and angina presented with bilateral leg weakness was admitted to the hospital. The subject had also developed ascending paralysis over several days. She had no history of preceding infection or vaccination. The initial laboratory findings were as follows: CK 326 IU/L, CK-MB 15.3 ng/mL, troponin I 0.850 ng/mL, and NT-proBNP 3,352 pg/mL. Electrocardiography revealed no significant ST segment change, but transthoracic echocardiogram showed regional wall motion abnormalities in the left ventricle and enlargement of the left atrium. In the coronary angiography that was performed to rule out myocardial infarction, there was no significant coronary artery stenosis. During admission, intermittent atrial fibrillation was observed, and a daily dose of clopidogrel 75 mg was prescribed as a preventive medication. After nine days, the patient was referred to the Department of Neurology. Neurologic examination revealed paralytic weakness in four extremities (MRC grade, right/left: arm 2/3, leg 1/1) and deep tendon reflexes were absent. A Foley catheter was inserted due to acute urinary retention. An initial nerve conduction study showed motor-dominant demyelinating polyneuropathy. GBS was highly suspected and IVIg 400 mg/kg/day was administered. Three days after IVIg infusion, abrupt onset of right hemiparesis and left gaze preponderance occurred. An MRI of the brain showed multiple embolic infarctions, mainly involving the left anterior cerebral artery territory (Fig. 2). Electrocardiography showed sinus tachycardia with a heart rate > 140 bpm. TEC associated with IVIg infusion were suspected, and thus IVIg therapy was discontinued by switching clopidogrel to edoxaban 30 mg daily. With active physical therapy and conservative management, the patient was discharged with improved motor performance (MRC grade, right/left; arm 2/4, leg 2/4).

4. Discussion

For over a two-year period in a single tertiary hospital, we identified 3 TEC cases that occurred during or after the completion of IVIg therapy. The rates of TEC per patient, per treatment course, and per infusion were 4.91% (3/61), 3.57% (3/84), and 0.82% (3/364), respectively. All patients with TEC were older than 60 years and had vascular risk factors such as hypertension, diabetes mellitus, or a history of stroke. Of note, in patients suffering from MG, the TEC associated with IVIg developed in 33.3% (2/6) of the patients, 25.0% (2/8) of the treatment courses, and 5.0% (2/40) of the infusions. Based on our experience and observations from previous studies, although the risk of developing TEC associated with IVIg is generally low in immune-mediated neurological disorders, we suggest that aged patients at high risk of thrombosis undergoing an aggressive course of disease such as an MG crisis require special attention to TEC when receiving IVIg therapy.

Since the first report of IVIg-related TEC was described in 1986 [10], several studies have attempted to determine the frequency and risk factors of TEC occurring in association with IVIg infusion in patients with various neurological, immunological, and hematological disorders [4–9,11–13]. The incidence of TEC in immune-mediated neurological disorders was relatively low in two prospective studies, ranging from 1.2% (per treatment course) [4] to 1.7% (per patient) [5]. It was reported as 3% (per patient) in another retrospective study [6]. In the context of the risk factors

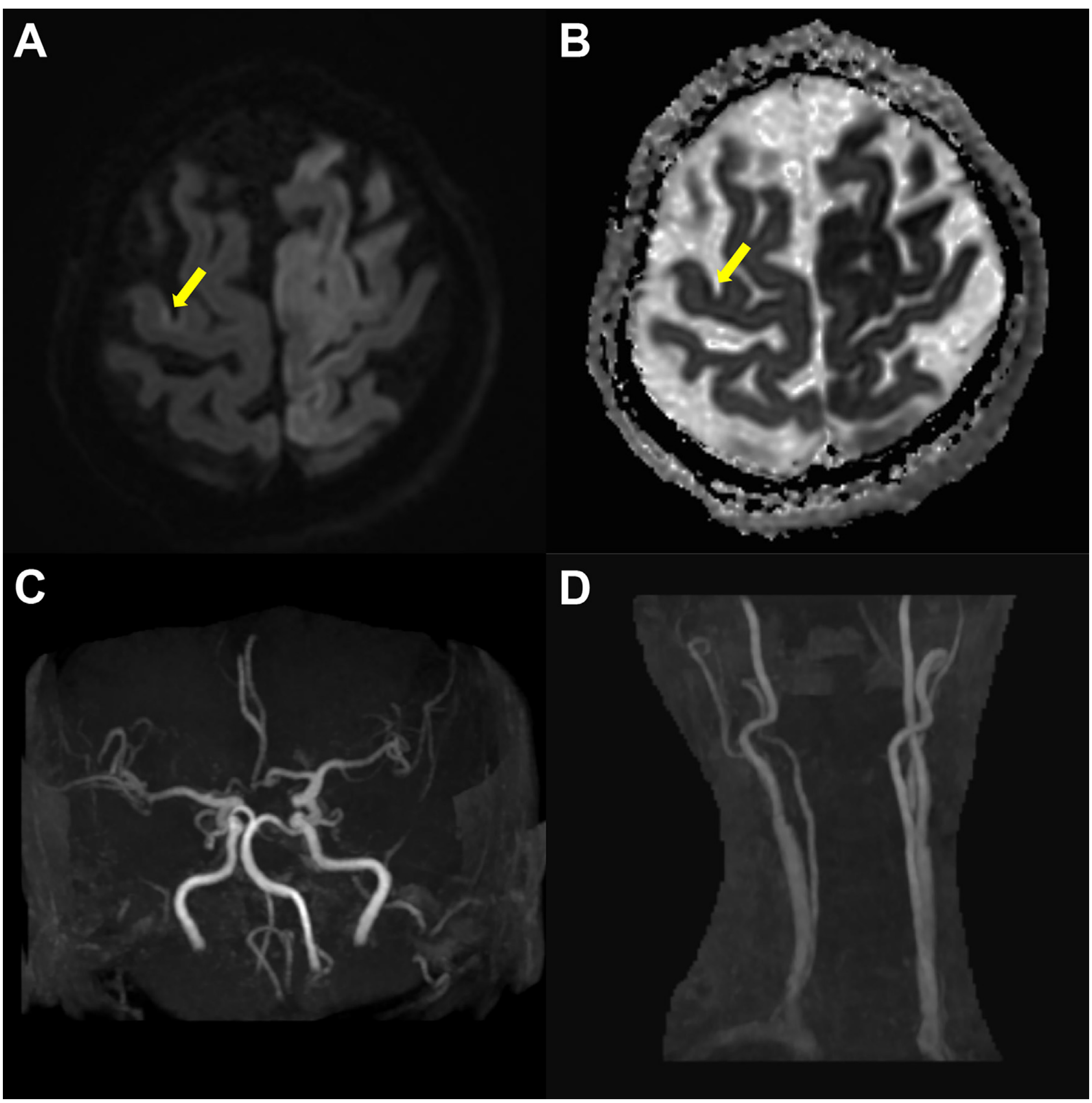

Fig. 2. Brain MRI of patient 3 (F/79, Guillain-Barré syndrome). Brain MRI (diffusion weighted image/apparent diffusion coefficient) shows acute infarction in left frontoparietal lobe. The arrows indicate another diffusion-restricted lesion in right precentral gyrus (A and B). Brain MRA shows left distal middle cerebral artery stenosis (C) and right carotid bulb stenosis (D).

of TEC, a temporal relationship between IVIg infusion and the occurrence of TEC was addressed in a large cohort study involving 10,759 patients with hematological malignancies [9]. IVIg might increase the serum viscosity and impair blood flow, thereby precipitating TEC [6,14]. Therefore, the incidence of TEC per infusion would be more informative than that per patient or treatment course, which was calculated as 0.82% in our study. Cerebral artery vasospasm and infusion of vasoactive cytokines or clotting factors can also be considered as the possible changes manifested upon the development of IVIg-associated TEC [15,16].

To our interest, two out of the six patients (of eight treatment courses) with MG, who had received IVIg therapy, developed TEC in this study. It is not known whether patients with MG are more prone to thrombosis than those with other neuromuscular disorders. In *patient 1*, IVIg was administered in the acute to subacute period of ischemic stroke and might have had potentiated thromboembolism. Moreover, both *patients 1 and 2* were critically ill with limited mobility (modified Rankin Scale scores 5 and 3, respectively), that facilitated the development of venous thrombosis. Hospitalized MG patients may have a two-fold increased risk of subsequent venous thromboembolism [17]. Hypercapnia caused by respiratory muscle weakness might also have contributed to increased blood viscosity [18]. Although the presence of right-to-left shunt was not examined in our patients, these risk factors might have potentiated arterial TEC such as cerebral infarction and pulmonary thromboembolism.

In the present study, all three patients were elderly women with vascular risk factors and may have been prone to thrombosis even in the absence of IVIg therapy. In particular, *patients 1 and 3* had suffered from stress cardiomyopathy prior to the initiation of IVIg infusion. Stress cardiomyopathy is characterized by reversible systolic and diastolic left ventricular dysfunction with variable regional wall motion abnormalities [19]. It predominantly affects elderly women with neurological or psychiatric disorders, and can be triggered by physical or emotional stimuli [19]. The clinical course of stress cardiomyopathy is generally benign and transient [19], but some critically ill patients, such as those in the advanced stages of amyotrophic lateral sclerosis, may experience fatal outcomes [20]. As described in previous reports [21], cerebral infarction might have arisen from stress cardiomyopathy in these two patients.

IVIg-associated risk factors of TEC include a high daily dose of IVIg, rapid infusion rate, and probably IVIG-naïve patients [7]. Our patients were all IVIg-naïve and were prescribed IVIg per the same standard protocol (400 mg/kg daily for five consecutive days). Although it was not apparent in our study, several previous studies provided the following justifications for the relationship between TEC and IVIg: (1) occurrence of events in younger, low-risk patients [22,23]; (2) strong temporal relationship including events during the infusion [22,23]; (3) strong association between the product brand and increased rates of thrombotic events vs. the control groups [24]; and (4) association of thrombotic events with thrombogenic impurities found in certain IVIg products and/-batches [25].

It is important to acknowledge the limitations of this study. Given the high risk of thrombosis in our patients, the occurrence of TEC after IVIg infusion may be a coincidental finding. Unfortunately, as this was a retrospective descriptive study with a small sample size, we could not analyze the risk factors associated with TEC in comparison with the IVIg-treated patients vs. the control group. Nonetheless, a recent larger study showed that the incidence of arterial and venous TECs was greater in the regularly IVIg-treated patients with inflammatory neuropathy than the population-based controls [26]. Vascular risk factors were more common in the TEC group, but there were no significant factors directly related to IVIg administration [26]. These results suggest that IVIg may have a small but clear contribution to TEC. In this sense, as shown in our patients, severe adverse events such as IVIg-related TEC, are rare but are more likely to occur in the most vulnerable high-risk groups. Further prospective studies with a larger number of patients are warranted to support the need for primary prevention in these patients.

In summary, we showed that the risk of developing TEC associated with IVIg is generally low in patients with immune-mediated neurological disorders. However, the frequency of TEC related to IVIg might not be negligible in critically ill elderly patients with vascular risk factors, and should therefore be cautiously monitored, especially in those suffering from MG crisis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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